

Hate speech on social media platforms

Guest lecture April 16th

Who am I?

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Anne Clausen is a PhD student at the Digital Democracy Centre at SDU. She holds a master degree in Public Health from Aarhus University. Her research focuses on investigating the spreading of intolerance on social media platforms. She is specially interested in investigating the dynamics of intolerance with machine learning methods.



Today's lecture

1st half:

- ☐ Theoretical background on hate speech and the opportunity structures social media platforms provide for its spread

Discussion with peers (5 mins):

- ☐ Why is hate speech on social media platforms a problem?
- ☐ Post your answers in Padlet
- ☐ Recap in plenum

2nd half:

- ☐ Insights from an ongoing study of hate speech targeting sexual and gender minority targets in Danish Facebook comments

What is hate speech?

- ❑ Discriminatory language or hateful expressions aimed at individuals because of their (ascribed) collective identity or membership in a social group (e.g. sexual orientation, gender identity, ethnicity, religion etc.) (United Nations, 2019; Frischlich, 2023)

Explicit hate speech

- ❑ Overt slurs, pejoratives or threats of violence (Paasch-Colberg et al., 2021)

Implicit hate speech

- ❑ Negative stereotyping (Rieger et al., 2021) or disguised as humor (Matamoros-Fernández et al., 2023)

References:

- Frischlich, L. (2023). Hate and harm. In S. Paasch-Colberg, C. Strippel, & M. Emmer (Eds.), *Challenges and perspectives of hate speech analysis: An interdisciplinary anthology* (Vol. 12). Digital Communication Research.
- Matamoros-Fernández, A., Bartolo, L., & Troynar, L. (2023). Humour as an online safety issue: Exploring solutions to help platforms better address this form of expression. *Internet Policy Review*, 12(1). <https://doi.org/https://doi.org/10.14763/2023.1.1677>
- Paasch-Colberg, S., Strippel, C., Trebbe, J., & Emmer, M. (2021). From Insult to Hate Speech: Mapping Offensive Language in German User Comments on Immigration. *Media and Communication*, 9(1), 171–180. <https://doi.org/https://doi.org/10.17645/mac.v9i1.3399>
- Rieger, D., Kümpel, A. S., Wich, M., Kiening, T., & Groh, G. (2021). Assessing the extent and types of hate speech in fringe communities: A case study of alt-right communities on 8chan, 4chan, and Reddit. *Social Media + Society*, 1–14. <https://doi.org/https://doi.org/10.1177/20563051211052906>
- United Nations. (2019). *The UN Strategy and Plan of Action on Hate Speech*. https://www.un.org/en/genocideprevention/documents/advising-and-mobilizing/Action_plan_on_hate_speech_EN.pdf

Hate speech on social media platforms

- ❑ Many social media platforms are available to users today, each characterized by distinct platform design (Bayer et al. 2020).
 - ❑ Mainstream platforms: Facebook, Instagram, YouTube, TikTok, Twitter/X
 - ❑ Alternative platforms: Telegram, Gab, 4chan
- ❑ The average user nowadays uses 6.8 different platforms each month (Kemp, 2025).
- ❑ Social media platforms differ from each other in terms of content modality (text, visual, audiovisual), technical features (e.g. like button), and affordances (e.g. anonymity).
- ❑ Different platform designs create different **opportunity structures** for the spreading of intolerance and hate speech in online spaces.

References:

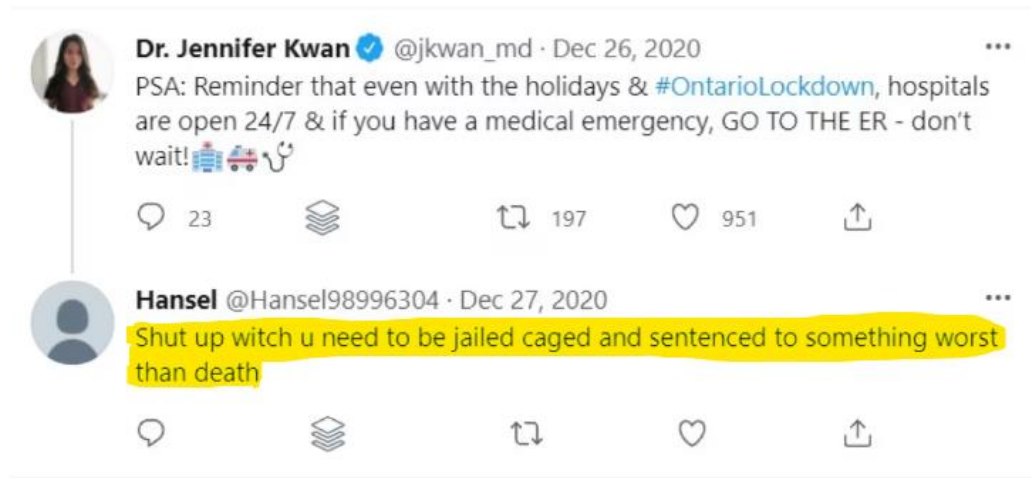
Bayer, J. B., Triêu, P., & Ellison, N. B. (2020). Social media elements, ecologies, and effects. *Annual Review of Psychology*, 71(1), 471–497. <https://doi.org/10.1146/annurev-psych-010419-050944>

Kemp, S. (2025). Digital 2025: Global overview report (p. 372). DATAREPORTAL. <https://datareportal.com/reports/digital-2025-global-overview-report>

Opportunity structures: content modalities

Hate speech on social media can be expressed in different content modalities:

- ☐ Textual (comments, messages etc.)
- ☐ Visual (images, memes etc.)
- ☐ Audiovisual (videos, reels etc.)



Robert Miller commented on your video



Feeling Like a Champion!



Robert Miller

Gross faggot. Fuck you. Go die of aids. Sodom and Gomorrah will return. Sick nasty pedo-fag

[REPLY](#) [MANAGE ALL COMMENTS](#)

Opportunity structures: content modalities

Multimodality:

- ❑ The combination of multiple modalities to express meaning (Adami, 2017)
- ❑ Social media content can be multimodal if it combines text, images, audio and video
 - ❑ Example: TikTok video with audio and text overlays
 - ❑ Example: Memes (images with text-embeddings)

References:

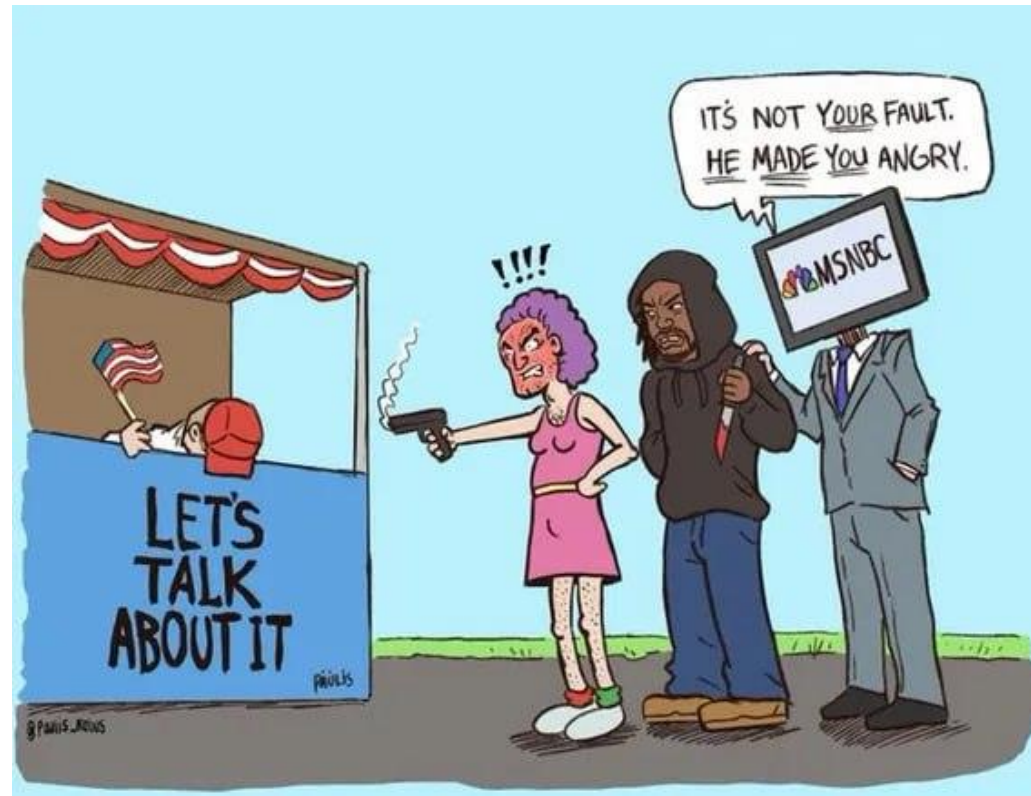
Adami, E. (2017). Introducing multimodality. In García O., Flores N., & Spotti M. (Eds.), *The Oxford handbook of language and society* (pp. 451–472). Oxford University Press.

Multimodal hate: memetic examples collected from public Telegram channels

NO WATER, BUT IT SURE IS GAY AF



Multimodal hate: memetic examples collected from public Telegram channels



Opportunity structures: platform features & affordances

Platform features (Bimber & Gil de Zúñiga, 2020; Schulze et al.2024):

- ❑ Technically embedded in the platform design – allow for user interaction with the content – examples: follow, share, comment, like.

Platform affordances (Evans et al., 2017; Schulze et al.2024):

- ❑ "Action possibilities" – emerges when users interact with the features of a platform – examples: anonymity, visibility, collectivity, interactivity

References:

- Bimber, B., & Gil de Zúñiga, H. (2020). The unedited public sphere. *New Media & Society*, 22(4), 700–715. <https://doi.org/10/gg8hmr>
- Evans, S. K., Pearce, K. E., Vitak, J., & Treem, J. W. (2017). Explicating affordances: A conceptual framework for understanding affordances in communication research. *Journal of Computer-Mediated Communication*, 22(1), 35–52. <https://doi.org/10.1111/jcc4.12180>
- Schulze, H., Greipl, S., Hohner, J., & Rieger, D. (2024). Social media and radicalization: An affordance approach for cross-platform comparison. *Medien & Kommunikationswissenschaft*, 72(2), 187–212. <https://doi.org/10.5771/1615-634X-2024-2-187>

Opportunity structures: platform features & affordances

Examples from Twitter/X and how it relates to the creation and spread of hate speech:

- ☐ Feature: freely selectable handle and account name
- ☐ Affordance: anonymity
 - ☐ Anonymity eases hate speech by allowing users to disconnect online behavior from offline identity, which can lead to toxic disinhibition and the expression of hatred unlikely to occur in face-to-face interactions (see online disinhibition effect, Suler, 2004)
- ☐ Feature: retweet/repost function
- ☐ Affordance: visibility
 - ☐ Hate speech achieves higher visibility than non-hateful content via social media sharing mechanisms, evidenced by retweet dynamics (Maarouf et al., 2024)

References:

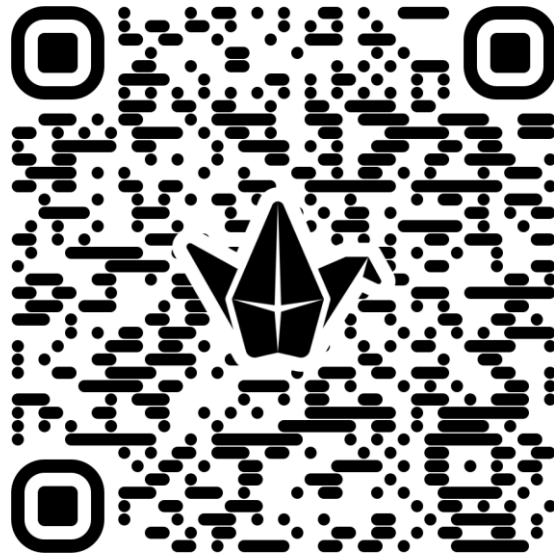
- Maarouf, A., Pröllochs, N., & Feuerriegel, S. (2024). The virality of hate speech on social media. *Proceedings of the ACM on Human-Computer Interaction*, 8(CSCW1), 1–22. <https://doi.org/10.1145/3641025>
- Suler, J. (2004). The online disinhibition effect. *CyberPsychology & Behavior*, 7(3), 321–326. <https://doi.org/10.1089/1094931041291295>

Discuss (5 mins): Why is hate speech on social media problematic?

Enter your answers into Padlet:

<https://padlet.com/aclausen11/why-is-hate-speech-on-social-media-problematic-4av06aut6vpa0vmd>

Recap in plenum in 5 mins



Why is hate speech on social media problematic?

- ❑ Hate speech is harmful to its victims and can trigger trauma responses in targets (Leets, 2002)
- ❑ Online hate speech incites to offline hate crime/violence (Arcila Calderón et al., 2024)
- ❑ Online hate speech is a threat to democratic participation because targets refrain from participating in online debates to avoid attacks, thereby affecting the inclusiveness of digital public spheres (Amnesty International Danmark, 2025)

References:

Amnesty International Danmark. (2025). *HAD SKADER – 22 ÅRS RETSPRAKSIS PÅ STRAFFELOVENS § 266 B*. https://amnesty.dk/wp-content/uploads/2025/01/Amnesty_HadskaderRapport2024_Digital_FINAL.pdf

Arcila Calderón, C., Sánchez Holgado, P., Gómez, J. *et al.* From online hate speech to offline hate crime: the role of inflammatory language in forecasting violence against migrant and LGBT communities. *Humanit Soc Sci Commun* **11**, 1369 (2024). <https://doi.org/10.1057/s41599-024-03899-1>

Leets, L. (2002). Experiencing hate speech: Perceptions and responses to anti-Semitism and antigay speech. *Journal of Social Issues*, 58(2), 341–361. <https://doi.org/10.1111/1540-4560.00264>

Insights from a study of hate speech in Danish Facebook comments

- ❑ The study examines content characteristics of explicit and implicit forms of hate speech targeting sexual and gender minorities in Facebook comments collected from public pages of Danish news media, politicians, political parties and citizen-driven groups.
- ❑ The study received a data donation from the NGO 'Os & Data' who originally collected the data for the report 'Attacks and Hate in the Public Debate on Facebook' (Analyse & Tal & Os & Data, 2025)
- ❑ The dataset covered years 2021-2024 and was collected end-year of 2024, and the data is therefore considered after moderation

References

Analyse & Tal, & Os & Data. (2025). Angreb & had i den offentlige debat på Facebook. <https://www.tryghed.dk/-/media/files/pdf/publikationer/2025/2025-angreb-og-had-i-den-offentlige-debat.pdf>



Insights from a study of hate speech in Danish Facebook comments

- ❑ The NGO used an AI-based algorithm (Att&ck) to identify attacks (broad definition including both offensive language and hate speech) and a keyword-based topic model to identify attacks targeting sexual and gender minorities (Analyse & Tal & Os & Data, 2025).
- ❑ The Att&ck algorithm identified 27.587 comments as attacks against sexual and gender minorities.
- ❑ The current study analyses a random sample of 2.000 comments

References

Analyse & Tal, & Os & Data. (2025). Angreb & had i den offentlige debat på Facebook. <https://www.tryghed.dk/-/media/files/pdf/publikationer/2025/2025-angreb-og-had-i-den-offentlige-debat.pdf>

Explicit and implicit hate speech in Danish Facebook comments

Manual content analysis of random sample of 2.000 comments

- ☐ The codebook was developed deductively based on dimensions highlighted as central to the characterization of explicit and implicit hate speech in previous research.
- ☐ Two human coders read through the comments and annotated for each comment if it contained characteristics of hate speech.

Explicit and implicit hate speech in Danish Facebook comments

Comments were coded as **explicit hate speech** if they used slurs, pejoratives or called for violence to attack the target group

- ☐ Slurs were considered as derogatory terms whose meaning is inherently discriminating against the target group (Hess, 2021) (e.g. the homophobic slur *'faggot'*)
- ☐ Pejoratives were considered descriptive words or expressions containing a negative evaluation about the target group (Miscevic, 2017), which does not in itself demean the target group (e.g. *'sick'*, *'stupid'*, *'disgusting'*)
- ☐ Calls for violence considered expressions of violence, threats, harm or killing (Paasch-Colberg et al., 2021)

References:

- Hess, L. (2021). Slurs: Semantic and Pragmatic Theories of Meaning. In P. Stalmaszczyk (Ed.), *The Cambridge Handbook of the Philosophy of Language* (pp. 450–466). Cambridge University Press. Cambridge Core. <https://doi.org/10.1017/9781108698283.026>
- Miscevic, N. (2017). The semantic structure of Pejoratives. *Linguistik Online*, 82. <https://doi.org/10.13092/lo.82.3714>
- Paasch-Colberg, S., Strippel, C., Trebbe, J., & Emmer, M. (2021). From Insult to Hate Speech: Mapping Offensive Language in German User Comments on Immigration. *Media and Communication*, 9(1), 171–180. <https://doi.org/https://doi.org/10.17645/mac.v9i1.3399>

Explicit and implicit hate speech in Danish Facebook comments

Comments were coded as *implicit hate* speech if they used negative stereotyping, humor or expressed limitation of rights to attack the target group.

- ❑ Negative stereotyping considered expressions of negative overgeneralizations or prejudice about the attributes or behaviors of the target group (Marx & Ko, 2019).
- ❑ Humor considered expressions of irony, sarcasm or jokes to demean the target group (Matamoros-Fernández et al., 2023).
- ❑ Expressions of limitation of rights considered mentions of removal of rights, freedoms and opportunities for the target group (ElSherief et al., 2021) (e.g. statements denying sexual minorities the right to family formation).

References:

- ElSherief, M., Ziems, C., Muchlinski, D., Anupindi, V., Seybolt, J., De Choudhury, M., & Yang, D. (2021). Latent Hatred: A Benchmark for Understanding Implicit Hate Speech. In M.-F. Moens, X. Huang, L. Specia, & S. W. Yih (Eds.), *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing* (pp. 345–363). Association for Computational Linguistics. <https://doi.org/10.18653/v1/2021.emnlp-main.29>
- Marx, D., & Ko, S. J. (2019). Stereotypes and Prejudice. *Oxford Research Encyclopedia of Psychology*. <https://doi.org/10.1093/acrefore/9780190236557.013.307>
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Explicit and implicit hate speech in Danish Facebook comments

Manual content analysis of random sample of 2.000 comments

- ☐ After manual coding 1,149 (57.5%) comments were identified as hate speech going against sexual or gender minorities
- ☐ Considerable co-occurrence between explicit and implicit types of hate speech in the dataset

Hate speech comments	1,149
Explicit hate speech*	882 (76.8%)
Slur	689 (60.0%)
Pejorative	382 (33.2%)
Calls for violence	29 (2.5%)
Implicit hate speech*	1,017 (88.5%)
Negative stereotyping	883 (76.8%)
Humor	254 (22.1%)
Limitation of rights	75 (6.5%)
Explicit only	132 (11.5%)
Implicit only	267 (23.2%)
*Explicit and implicit hate speech comments are non-mutually exclusive, and comments can contain characteristics of both	

Explicit and implicit hate speech in Danish Facebook comments

- ☐ Strong positive correlation between Explicit and Slur
- ☐ Strong positive correlation between Implicit and Negative stereotyping
- ☐ Explicit and implicit hate speech frequently overlap rather than occur in isolation



Content characteristics of explicit and implicit hate speech in Danish Facebook comments

- Lexical characteristics of the different types of hate speech were examined through tf-idf (term frequency-inverse document frequency), given by (Manning et al., 2008):

$$\text{tf-idf}_{t,d} = \text{tf}_{t,d} \times \text{idf}_t$$

The term frequency ($\text{tf}_{t,d}$) measures how often a word appears in a document, given by:

$$\text{tf}_{t,d} = \frac{\text{number of times the word } t \text{ appears in document } d}{\text{total number of words in document } d}$$

The inverse document frequency (idf_t) measures the uniqueness of a word across a collection of documents, given by:

$$\text{idf}_t = \log \frac{N}{\text{df}_t}$$

References:

Manning, C. D., Raghavan, P., & Schütze, H. (2008). Scoring, term weighting and the vector space model. In *Introduction to Information Retrieval* (pp. 109–133). Cambridge University Press. (6).

Content characteristics of explicit and implicit hate speech in Danish Facebook comments

- ❑ The analysis identified the top 5 terms per hate speech according to tf-idf rank (11*5=55 terms=too many to show here)
- ❑ Most prominent tf-idf findings showcased here

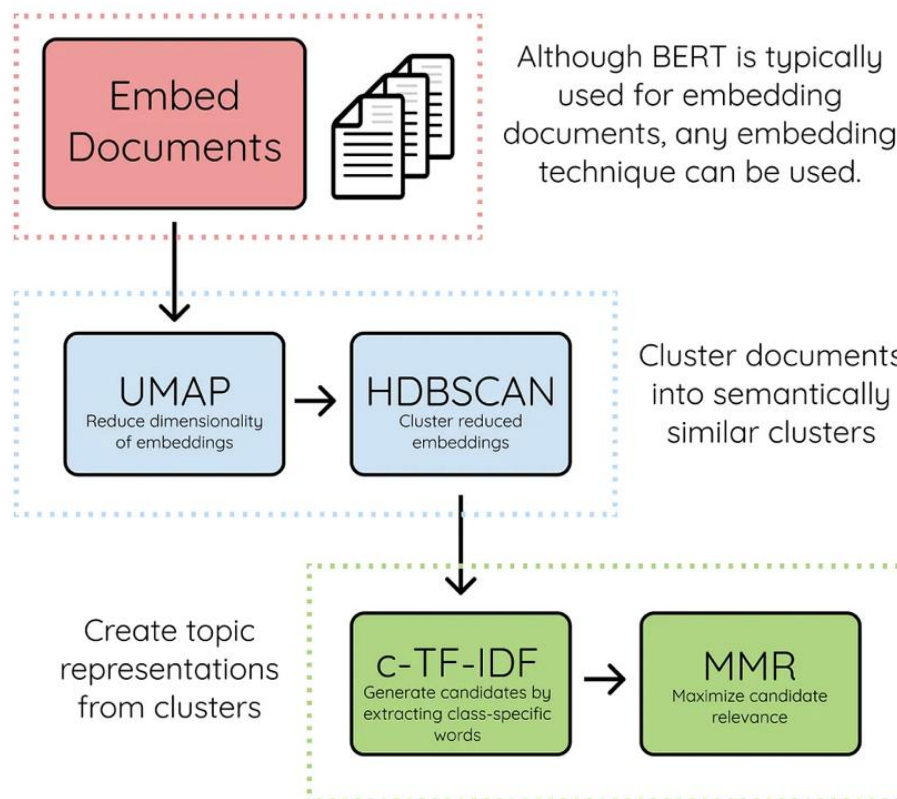
Tf-idf rank	Term (da/en)
Total hate speech comments (n=1,149)	
1	svans (faggot)
2	køn (gender/sex)
3	bøsse (gay/fag)
4	homo (homo)
5	bøsser (gays/fags)

Tf-idf rank	Term (da/en)
Explicit hate speech (n=882)	
1	svans (faggot)
Implicit hate speech (n=1,017)	
1	køn (gender/sex)

Content characteristics of explicit and implicit hate speech in Danish Facebook comments

- ❑ Thematic characteristics of the different types of hate speech were examined through BERTopic (Grootendorst, 2022)
- ❑ Embeddings capture semantic meaning, compared to other topic modeling models who rely on term frequency

References:
Grootendorst, M. (2022). *BERTopic: Neural topic modeling with a class-based TF-IDF procedure* (arXiv:2203.05794). arXiv.
<https://doi.org/10.48550/arXiv.2203.05794>



Content characteristics of explicit and implicit hate speech in Danish Facebook comments

❑ The analysis identified the top 5 topics per hate speech type (11*5=55 topics=too many to show here)

❑ Most prominent BERTopic findings for **explicit hate speech** (n=882) showcased here:

Topic number	Topic label	Topic representations
1	Homosexuality debates	bøsse, tror, andres, mener, homoseksuelle, hvorfor, synes, hver, heller, mennesker (gays/fags, think, others, believe, homosexuals, why, think, each, either, people)
2	Homosexuality debates	bøsse, homoseksuel, lgbt, bøsser, nogensinde, tak, kun, eu, jeres, bare (gay/fag, homosexual, lgbt, gays/fags, ever, thanks, only, eu/european union, yours, just)
3	Disgust-based derogations	køn, nååå, bøsserøve, lillefinger, kom, kun, røvfuld, føj, går, skovl (gender/sex, ohhh, faggots, little finger, come, only, ass kicking, ew, walk, shovel)
4	Advocating physical assault	bøsse, hankøn, mener, afsky, nogensinde, næ, dræbes, køn, tænker, nogen (gay/fag, male gender/sex, believe, disgust, ever, no, kill, gender/sex, think, someone)
5	Moralizing insults	bøssekarlene, bøssekarl, bøsse, bøssevenner, bøssekoven, homo, gay, skod, satanisk, svans (faggots, faggot, gay/fag, gay/fag friends, gay/fag forest, homo, gay, shitty, satanic, faggot)

Content characteristics of explicit and implicit hate speech in Danish Facebook comments

□ Most prominent BERTopic findings for **call for violence** (n=29) showcased here:

Topic number	Topic label	Topic representations
1	Advocating physical assault	kneppet, kneppede, røvfuld , mener, laver, bøsserøve, skær , bøsserøv, kæft, kun (fucked, fucked, ass kicking , believe, do, ass faggots, cut , faggot, only)
2	LGBT-phobic slurs	røven, bøssen , bøsse , lorte, queer, lgbtqrstuvwxyzæøå , værste, skrev, sagde, stod (the ass, the gay/fag , gay/fag , shitty , queer , lgbtqxxxxxx , worst, wrote, said, stood)
3	Advocating physical assault	lgbtqia, sagde, forpulede, menneskers, lesbisk, smæk , mand, kvindes, smadre , sagt (lgbtqia, said, fucking, peoples', lesbian, spanking , man, womans', smash , said)
4	Moralizing insults	synd , bare, svans, spjældet , ser, bøssekarl, kan, nosseme, hænge, mand (sin , just, faggot, jail , see, faggot, can, balls, hang, man)
5	Advocating physical assault	fanden, dræbes , uanset, fucking, fortabt, bøsse, må, luder, kommer, helst (the devil, kill , no matter, fucking, lost, gay/fag, can, whore, come, rather)

Content characteristics of explicit and implicit hate speech in Danish Facebook comments

□ Most prominent BERTopic findings for *implicit hate speech* (n=1,017) showcased here:

Topic number	Topic label	Topic representations
1	Homosexuality debates	bøsser, bøsse, mener, hver, synes, tror, siger, hvorfor, mennesker, lidt (gays/fags, gay/fag, believe, each, think, believe, says, why, people, little)
2	Gender identity debates	køn, hunkøn, uanset, mennesker, mener, sige, tænker, andet, kun, eksistere (gender/sex, female gender/sex, no matter, people, believe, say, thinks, other, only exist)
3	Mocking LGBT+ identities	bøsse, køn, bøsser, sige, fuck, bøsserøve, uanset, seksualitet, ladyboys, ladyboy (gay/fag, gender/sex, gays/fags, say, fuck, ass faggots, no matter, sexuality, ladyboys, ladyboy)
4	Gender diversity hostility	køn, bøsser, mand, mænd, mennesker, personen, helst, fjols, sige, kun (gender/sex, gays/fags, man, men, people, the person, rather, fool, say, only)
5	Gender identity debates	køn, hankøn, virkelig, vedkommende, kønsskifte, mener, hvorfor, andet, alligevel, samt (gender/sex, male gender/sex, really, the person, gender/sex change, believes, why, other, anyway, also)

Content characteristics of explicit and implicit hate speech in Danish Facebook comments

□ Most prominent BERTopic findings for *limitation of rights* (n=75) showcased here:

Topic number	Topic label	Topic representations
1	Gender segregation debates	kønsskifte, køn, mandefængsel, mændene, kvindefængsel, mænd, vedkommende, transvestit, hankøn, penis (gender/sex change, gender/sex, male prison, the men, female prison, men, the person, transvestite, male gender/sex, penis)
2	Pathologizing sexual minorities	mens, bøssedreng, lgbtq, simpelthen, homo, vrangforestillinger, bør, fanden, røven, sagt (while, gay/fag boy, lgbtq, simply, homo, delusions, should, the devil, the ass, said)
3	Gender category differentiation	transkønnede, transkvinder, transpersoner, lgbt, køn, køns, trans, mands, kvinder, mænd (transgender, transwomen, transpeople, lgbt, gender/sex, gender/sex, trans, mens', women, men)
4	Concerns for children	bøsse, homoseksuel, lesbiske, køn, hvorfor, bøsser, dumme, barn, mennesker, voksne (gay/fag, homosexual, lesbians, gender/sex, why, gays/fags, stupidest, child, people, adults)
5	Concerns for children	køn, voksne, voksen, hvorfor, provokerende, forskellige, børn, seksualitet, meninger, årige (gender/sex, adults', adult, why, provoking, different, children, sexuality, opinions, year-old)

In sum:

Thematic split where explicit comments favor direct insults and mocking, while implicit comments are characterized by broader debates around gender identity and the marginalization of sexual and gender minority individuals

Thanks for listening

Any questions?